

$$1. \quad \frac{dP}{dt} = P - P^2 \quad (\text{separable})$$

$$\int \frac{dP}{P(1-P)} = \int dt$$

Partial Fractions

$$\frac{1}{P(1-P)} = \frac{A}{P} + \frac{B}{1-P}$$

$$\Rightarrow \begin{cases} A=1 \\ B=1 \end{cases}$$

$$\int \left(\frac{1}{P} + \frac{1}{1-P} \right) dP = \int dt$$

$$\ln|P| - \ln|1-P| = t + C_1$$

$$\ln \left| \frac{P}{1-P} \right| = t + C_1$$

$$\left| \frac{P}{1-P} \right| = e^t e^{C_1}$$

$$\Rightarrow \frac{P}{1-P} = \underbrace{(\pm e^{C_1})}_C e^t = C e^t$$

Since $P(0) = 2$, then $\frac{2}{1-2} = C e^0 \rightarrow C = -2$

$$\frac{P}{1-P} = -2e^t \rightarrow P = -2e^t + 2Pe^t \rightarrow 2e^t = P(2e^t - 1)$$

$$\rightarrow \boxed{P = \frac{2e^t}{2e^t - 1}}$$

2. We have a linear DE.

$$\text{Standard form: } \frac{dy}{dx} + \underbrace{\frac{4}{x}}_{P(x)} y = \frac{\cos(3x)}{x^3} \quad (*)$$

$$\text{Int. factor: } e^{\int P(x) dx} = e^{\int \frac{4}{x} dx} = x^4$$

$$\text{Multiply } (*) \text{ by } x^4 \text{ to get: } x^4 \frac{dy}{dx} + 4x^3 y = x \cos(3x)$$

$$\frac{d}{dx} [x^4 y] = x \cos(3x)$$

$$\rightarrow x^4 y = \int x \cos(3x) dx$$

$$\rightarrow \boxed{x^4 y = \frac{x}{3} \sin(3x) + \frac{1}{9} \sin(3x) + C}$$

3. Standard form: $y' + \frac{2}{x} y = 3 \quad (*)$

$$\text{int. factor: } e^{\int \frac{2}{x} dx} = x^2 \rightarrow x^2 y' + 2xy = 3x^2$$

$$\Rightarrow (x^2 y)' = 3x^2 \xrightarrow{\text{integrate}} x^2 y = x^3 + C$$

$$y = x + \frac{C}{x^2}$$

$$y(1) = 0 \rightarrow 0 = 1 + C \rightarrow C = -1$$

$$\therefore \boxed{y = x - \frac{1}{x^2}}$$

4. $\frac{dy}{dx} = \frac{y^2}{xy+x^2}$ Homogeneous: $y = vx$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Substitute to get: $v + x \frac{dv}{dx} = \frac{v^2}{v+1}$ [separable]

$$x \frac{dv}{dx} = \frac{v^2 - v(v+1)}{v+1} = -\frac{v}{v+1}$$

$$\Rightarrow \int \frac{v+1}{v} dv = -\int \frac{1}{x} dx \Rightarrow v + \ln|v| = -\ln|x| + C$$

$$\Rightarrow v + \ln|vx| = C \Rightarrow \frac{y}{x} + \ln|y| = C$$

or $y + x \ln|y| = Cx$

5. $\frac{dy}{dx} = \frac{x+3y}{3x+y}$ Homogeneous: $y = vx$

$$\rightarrow v + x \frac{dv}{dx} = \frac{1+3v}{3+v} \rightarrow x \frac{dv}{dx} = \frac{1+3v - v(3+v)}{3+v}$$

$$\rightarrow x \frac{dv}{dx} = \frac{1-v^2}{3+v} \rightarrow \int \frac{3+v}{1-v^2} dv = \int \frac{dx}{x}$$

P.F.

$$\frac{3+v}{(1+v)(1-v)} = \frac{A}{1+v} + \frac{B}{1-v}$$

$$\begin{cases} A = 1 \\ B = 2 \end{cases}$$

$$\int \left(\frac{1}{1+v} + \frac{2}{1-v} \right) dv = \int \frac{1}{x} dx$$

$$\ln|1+v| - 2\ln|1-v| = \ln|x| + C$$

$$\ln \left| \frac{1+v}{(1-v)^2} \right| = \ln|x| + C$$

$$\Rightarrow \ln \left| \frac{1+y/x}{(1-y/x)^2} \right| = \ln|x| + C$$

6. $xy' - y = 2y^3$ [Bernoulli with $n=3$]

The DE is also separable but we solve it as a Bernoulli.

$$y' - \frac{y}{x} = \frac{2y^3}{x}, \quad \text{let } u = y^{1-n} = y^{-2},$$

$$\rightarrow u' = -2y^{-3} y', \quad \text{Multiply the DE by } -2y^{-3}.$$

$$\rightarrow -2y^{-3} y' + \frac{2y^{-2}}{x} = -\frac{4}{x}$$

$$\rightarrow u' + \frac{2}{x} u = -\frac{4}{x} \quad (\text{Linear})$$

$$\text{Int. factor: } e^{\int \frac{2}{x} dx} = x^2$$

$$\rightarrow (ux^2)' = -4x$$

$$\rightarrow ux^2 = -2x^2 + C$$

$$\rightarrow u = -2 + \frac{C}{x^2}$$

$$\rightarrow \boxed{y^{-2} = -2 + \frac{C}{x^2}}$$

6. method (2)

$$x \frac{dy}{dx} = y + 2y^3 \quad \leftarrow \text{separable}$$

$$\frac{dy}{y(1+2y^2)} = \frac{dx}{x}$$

$$\begin{array}{c} \text{P.F.} \\ \frac{1}{y(1+2y^2)} = \frac{A}{y} + \frac{By+C}{1+2y^2} \end{array}$$

$$\rightarrow \begin{cases} A = 1 \\ B = -2 \\ C = 0 \end{cases}$$

$$\int \left(\frac{1}{y} - \frac{2y}{1+2y^2} \right) dy = \int \frac{dx}{x}$$

$$\ln|y| - \frac{1}{2} \ln(1+2y^2) + C_1 = \ln|x|$$

$$2 \ln|y| - \ln(1+2y^2) + 2C_1 = 2 \ln|x|$$

$$\Rightarrow 2C_1 = \ln \left(\frac{x^2(1+2y^2)}{y^2} \right)$$

$$\Rightarrow e^{2C_1} = \frac{x^2(1+2y^2)}{y^2}$$

$$\Rightarrow \frac{C}{x^2} = \frac{1}{y^2} + 2 \Rightarrow \boxed{y^{-2} = -2 + Cx^{-2}}$$

$$7. \quad y' = xy^4 - y \rightarrow y' + y = xy^4$$

Bernoulli with $n=4$. Let $u = y^{-3}$

$$\rightarrow u' = -3y^{-4}y'$$

Multiply the DE by $-3y^{-4}$ to get:

$$-3y^{-4}y' - 3y^{-3} = -3x$$

$$\rightarrow u' - 3u = -3x \quad [\text{linear}]$$

Int. factor: $e^{\int -3dx} = e^{-3x}$

$$\rightarrow (ue^{-3x})' = -3xe^{-3x}$$

$$ue^{-3x} = \int -3xe^{-3x} dx$$

$$ue^{-3x} = xe^{-3x} + \frac{1}{3}e^{-3x} + C_1$$

D	I
$-3x \oplus$	e^{-3x}
$-3 \ominus$	$-\frac{1}{3}e^{-3x}$
0	$\frac{1}{9}e^{-3x}$

$$u = x + \frac{1}{3} + C_1 e^{3x} \rightarrow 3u = 3x + 1 + C e^{3x}$$

$$\frac{3}{y^3} = 3x + 1 + C e^{3x} \rightarrow$$

$$y = \sqrt[3]{\frac{3}{3x + 1 + C e^{3x}}}$$

$$8. \quad x \frac{dy}{dx} + y = \frac{e^{2x}}{xy^2} \rightarrow \frac{dy}{dx} + \frac{y}{x} = \frac{e^{2x}}{x^2} y^{-2}$$

Bernoulli with $n = -2 \rightarrow u = y^{1-n} = y^3$

$$\frac{du}{dx} = 3y^2 \frac{dy}{dx} \quad (\text{Multiply the DE by } 3y^2)$$

$$\rightarrow 3y^2 \frac{dy}{dx} + \frac{3y^3}{x} = \frac{3e^{2x}}{x^2}$$

$$\rightarrow \frac{du}{dx} + \frac{3}{x} u = \frac{3e^{2x}}{x^2} \quad (\text{Linear})$$

Int. factor : $e^{\int \frac{3}{x} dx} = x^3$

$$\rightarrow \frac{d}{dx} \left[u x^3 \right] = 3x e^{2x}$$

$$\rightarrow u x^3 = \int 3x e^{2x} dx$$

$$\rightarrow u x^3 = \frac{3}{2} x e^{2x} - \frac{3}{4} e^{2x} + C_1$$

$$\rightarrow y^3 x^3 = \frac{3}{2} x e^{2x} - \frac{3}{4} e^{2x} + C_1$$

$$\rightarrow \boxed{4y^3 x^3 = (6x - 3) e^{2x} + C}$$

$$9. \underbrace{(y^2 \cos x - 3x^2 y - 2x)}_M dx + \underbrace{(ky \sin x - x^3 + 4y)}_N dy = 0$$

$$\begin{cases} \frac{\partial M}{\partial y} = 2y \cos x - 3x^2 \\ \frac{\partial N}{\partial x} = ky \cos x - 3x^2 \end{cases} \quad \text{if } k=2, \text{ the DE is exact.}$$

Assume $k=2$.

$$\frac{\partial f}{\partial x} = M = y^2 \cos x - 3x^2 y - 2x \xrightarrow{\text{Integrate}} f = y^2 \sin x - x^3 y - x^2 + g(y)$$

$$\frac{\partial f}{\partial y} = N = 2y \sin x - x^3 - 4y \xrightarrow{\text{Integrate}} f = y^2 \sin x - x^3 y - 2y^2 + h(x)$$

The solution of the DE is : $f = c$

$$\Rightarrow \boxed{y^2 \sin x - x^3 y - 2y^2 = c}$$

$$10. \frac{dy}{dx} = \frac{1 - (x+y)}{(x+y)}, \text{ let } u = x+y \rightarrow \frac{du}{dx} = 1 + \frac{dy}{dx}$$

$$\Rightarrow \frac{du}{dx} = 1 + \frac{1-u}{u} = \frac{1}{u}$$

$$\Rightarrow \int u du = \int dx$$

$$\Rightarrow \frac{u^2}{2} = x + C_1 \Rightarrow u^2 = 2x + C$$

$$\Rightarrow \boxed{(x+y)^2 = 2x + C}$$

$$11. \quad \frac{dy}{dx} = (4x+y)^2, \quad y(0)=2.$$

$$\text{Let } u = 4x+y \rightarrow \frac{du}{dx} = 4 + \frac{dy}{dx}$$

$$\rightarrow \frac{du}{dx} = 4 + u^2$$

$$\rightarrow \int \frac{1}{4+u^2} du = \int dx$$

$$\rightarrow \frac{1}{2} \operatorname{Arctan}\left(\frac{u}{2}\right) = x + C_1$$

$$\rightarrow \operatorname{Arctan}\left(\frac{u}{2}\right) = 2x + C$$

$$\rightarrow \operatorname{Arctan}\left(\frac{4x+y}{2}\right) = 2x + C$$

$$\text{Set } x=0 \text{ and } y=2, \text{ We get : } \operatorname{Arctan}(1) = 0 + C$$

$$\rightarrow C = \pi/4$$

$$\Rightarrow \frac{4x+y}{2} = \tan\left(2x + \pi/4\right)$$

$$\Rightarrow \boxed{y = 2 \tan\left(2x + \pi/4\right) - 4x}$$

$$12. \quad y' - \frac{1}{x} y = \frac{y^4}{x^2} \quad (*)$$

The DE is Bernoulli with $n=4$.

$$\text{Let } u = y^{-3} \rightarrow u' = -3y^{-4} y',$$

Multiply (*) by $-3y^{-4}$ to get:

$$-3y^{-4} y' + \frac{3}{x} y^{-3} = -\frac{3}{x^2}$$

$$\Rightarrow u' + \frac{3}{x} u = -\frac{3}{x^2} \quad (\text{Linear})$$

$$\text{Int. factor: } e^{\int \frac{3}{x} dx} = x^3$$

$$\Rightarrow (u x^3)' = -3x$$

$$\Rightarrow u x^3 = -\frac{3}{2} x^2 + C$$

$$\Rightarrow u = -\frac{3}{2x} + \frac{C}{x^3}$$

$$\Rightarrow \boxed{\frac{1}{y^3} = -\frac{3}{2x} + \frac{C}{x^3}}$$

13.

$$\underbrace{(1 + xy e^x)}_M dx + \underbrace{(x e^x + x \cos y)}_N dy = 0 \quad (*)$$

$$\frac{\partial M}{\partial y} = x e^x \quad \text{and} \quad \frac{\partial N}{\partial x} = e^x + x e^x + \cos y$$

The DE is not exact.

Let's see if we can find an integrating factor.

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{-e^x - \cos y}{x e^x + x \cos y} = -\frac{1}{x}$$

$\downarrow$
 Depends only
 on x .

$$\text{Int. factor: } e^{\int -\frac{1}{x} dx} = x^{-1}$$

Multiply (*) by $\frac{1}{x}$ to get an exact DE.

$$\left(\frac{1}{x} + y e^x \right) dx + (e^x + \cos y) dy = 0$$

$$f = \int \left(\frac{1}{x} + y e^x \right) dx = \ln|x| + y e^x + g(y)$$

$$F = \int (e^x + \cos y) dy = y e^x + \sin y + h(x)$$

$$\Rightarrow \boxed{\ln|x| + y e^x + \sin y = C}$$

14. Let $N = N(t)$ = # of bacteria at time t minutes.

$$\frac{dN}{dt} \propto N \Rightarrow \frac{dN}{dt} = k N$$

for a constant k

$$\int \frac{dN}{N} = \int k dt \rightarrow \ln N = kt + c \quad (N > 0)$$
$$\rightarrow N = e^c e^{kt}$$

$$@ t = 0, N(0) = N_0 \rightarrow N_0 = e^c$$

$$\rightarrow N = N_0 e^{kt}$$

We are given that :

t	N
0	100
20	500

$$N_0 = 100$$

$$\text{and } 500 = 100 e^{20k} \rightarrow k = \frac{1}{20} \ln(5)$$

Then, when $t = 30$ minutes we have :

$$N = 100 e^{30k}$$

$\approx$

1118 bacteria

$$15. \quad \frac{dm}{dt} = (\text{Rate in}) - (\text{Rate out})$$

$$\frac{dm}{dt} = (5 \text{ g/L}) (4 \text{ L/min}) - \left(\frac{m}{200} \text{ g/L} \right) (4 \text{ L/min})$$

$$\frac{dm}{dt} = 20 - \frac{m}{50}, \quad m(0) = 100$$

$$\frac{dm}{dt} + \frac{1}{50} m = 20 \quad [\text{Linear}]$$

$$\text{Int. factor: } e^{\int \frac{1}{50} dt} = e^{t/50}$$

$$\Rightarrow \frac{d}{dt} \left[m e^{t/50} \right] = 20 e^{t/50}$$

$$\Rightarrow m e^{t/50} = \frac{20 e^{t/50}}{(1/50)} + C$$

$$\Rightarrow m = 1000 + C e^{-t/50}$$

$$m(0) = 100 \rightarrow C = -900 \rightarrow (a) \boxed{m = 1000 - 900 e^{-t/50}}$$

$$(b) \quad \lim_{t \rightarrow \infty} m = 1000 \text{ g}$$

$$(c) \quad \text{Solve for } t, \quad m = 600 \text{ g to get: } \boxed{t = 40.5 \text{ min}}$$

$$16. \quad \frac{dT}{dt} + kT = kT_a \quad [\text{Linear}]$$

Assume T_a is constant.

(a) Int. factor: e^{kt}

$$\rightarrow \frac{d}{dt} [T e^{kt}] = kT_a e^{kt}$$

$$\rightarrow T e^{kt} = \int kT_a e^{kt} dt = T_a e^{kt} + C$$

$$\rightarrow T = T_a + C e^{-kt}$$

@ $t=0$, $T=T_0 \rightarrow C = T_0 - T_a$

$$\rightarrow T = T_a + (T_0 - T_a) e^{-kt}$$

(b) Assume t is in hours and

$t=0$ corresponds to 8:00 pm.

$$t=0 \rightarrow T = 25^\circ\text{C}$$

$$t=2 \rightarrow T = 15^\circ\text{C}$$

$$T_a = 10^\circ\text{C}$$

Find t when $T = 37^\circ\text{C}$

$$T = 10 + 15 e^{-kt}$$

$$t=2 \rightarrow 15 = 10 + 15 e^{-2k} \rightarrow k = -\frac{1}{2} \ln(1/3) = \frac{\ln(3)}{2}$$

$$37 = 10 + 15 e^{-kt} \rightarrow t = -\frac{1}{k} \ln\left(\frac{27}{15}\right) = -1.07 \text{ hrs}$$

$$t = -64 \text{ minutes}$$

$\rightarrow$ the murder took place 64 minutes before 8:00 pm, at 6:56 pm

$$17. \quad (a) \quad \frac{dv}{dt} + \frac{k}{m} v = g \quad [\text{Linear}]$$

$$\text{int. factor: } e^{\int \frac{k}{m} dt} = e^{kt/m}$$

$$\rightarrow \frac{d}{dt} \left[v e^{kt/m} \right] = g e^{kt/m}$$

$$\rightarrow v e^{kt/m} = \frac{mg}{k} e^{kt/m} + C$$

$$\rightarrow v = \frac{mg}{k} + C e^{-kt/m}$$

$$V(0) = V_0 \rightarrow C = V_0 - \frac{mg}{k}$$

$$(a) \quad V = \frac{mg}{k} + \left(V_0 - \frac{mg}{k} \right) e^{-kt/m}$$

$$(b) \quad \lim_{t \rightarrow \infty} V = \frac{mg}{k}$$

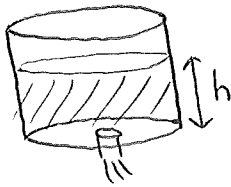
$$(c) \quad V_0 = 0, \quad m = 3 \text{ kg}, \quad V = 20 \text{ m/s}$$

$$g = 9.8 \text{ m/s}^2, \quad k = 0.5 \text{ N/(m/s)}$$

$$\text{solve for } t \text{ to get } t = -\frac{m}{k} \ln \left(1 - \frac{kV}{mg} \right)$$

$$t = 2.5 \text{ seconds}$$

18.



$$\frac{dh}{dt} = -\frac{a}{A} \sqrt{2gh}$$

(a)

$$\text{let } k = \frac{a}{A} \sqrt{2g}$$

$$\frac{dh}{dt} = -k \sqrt{h} \rightarrow \int \frac{dh}{\sqrt{h}} = - \int k dt$$

$$2\sqrt{h} = -kt + C_1 \rightarrow \sqrt{h} = C - \frac{kt}{2}$$

$$\text{@ } t=0, h=h_0 \Rightarrow C = \sqrt{h_0}$$

$$\therefore h = \left(\sqrt{h_0} - \frac{kt}{2} \right)^2$$

$$(b) \quad h=0 \rightarrow \sqrt{h_0} - \frac{kt}{2} = 0$$

$$\rightarrow t = \frac{2\sqrt{h_0}}{k}$$

TOP

$$R = 40 \text{ cm}$$

$$R = 0.4 \text{ m}$$

$$A = \pi R^2$$

Hole

$$r = 1.5 \text{ cm}$$

$$r = 0.015 \text{ m}$$

$$a = \pi r^2$$

$$k = \frac{a}{A} \sqrt{2g}$$

$$k = \frac{r^2}{R^2} \sqrt{2g}$$

Tank is full initially

$$h_0 = 3 \text{ m}$$

$$t = \frac{2\sqrt{h_0}}{k} = \frac{2R^2 \sqrt{h_0}}{r^2 \sqrt{2g}} = 556.4 \text{ seconds} = \boxed{9.3 \text{ minutes}}$$

$$19. \quad x y'' - (1+3x) y' + 3y = 0$$

$$y'' - \frac{(1+3x)}{x} y' + \frac{3}{x} y = 0 \quad [\text{standard form}]$$

$$P(x) = -\frac{(1+3x)}{x}, \quad y_1 = e^{3x} \text{ is a solution.}$$

$$\rightarrow y_2 = y_1 \int \frac{e^{-\int P(x) dx}}{y_1^2} dx$$

$$\rightarrow y_2 = e^{3x} \int \frac{e^{\int \frac{1+3x}{x} dx}}{(e^{3x})^2} dx$$

$$\rightarrow y_2 = e^{3x} \int \frac{e^{\ln x + 3x}}{e^{6x}} dx$$

$$\rightarrow y_2 = e^{3x} \int x e^{-3x} dx$$

$$\rightarrow y_2 = e^{3x} \left[-\frac{x}{3} e^{-3x} - \frac{1}{9} e^{-3x} \right]$$

$$\rightarrow y_2 = -\frac{x}{3} - \frac{1}{9}$$

D	I
x	$\oplus e^{-3x}$
1	$\ominus -\frac{1}{3} e^{-3x}$
0	$\frac{1}{9} e^{-3x}$

Scale it by a factor $(-9) \Rightarrow y_2 = 3x + 1$

$$\therefore \boxed{y = C_1 e^{3x} + C_2 (3x + 1)} \quad \text{General Solution.}$$

$$20. \begin{cases} y'' - 4x y' + (4x^2 - 2)y = 0 \\ y(0) = 1 \\ y'(0) = 2 \end{cases}$$

$y_1 = e^{x^2}$ is a solution of the DE.

$$y_2 = y_1 \int \frac{e^{-\int P(x) dx}}{y_1^2} dx$$

$$P(x) = -4x$$

$$\rightarrow y_2 = e^{x^2} \int \frac{e^{\int 4x dx}}{(e^{x^2})^2} dx$$

$$\rightarrow y_2 = e^{x^2} \int \frac{\cancel{e^{2x^2}}}{\cancel{e^{2x^2}}} dx = x e^{x^2}$$

$$y = c_1 y_1 + c_2 y_2 = (c_1 + c_2 x) e^{x^2}$$

$$y' = c_2 e^{x^2} + (c_1 + c_2 x) \cdot 2x e^{x^2}$$

$$y(0) = 1 \rightarrow c_1 = 1$$

$$y'(0) = 2 \rightarrow c_2 = 2$$

$$\therefore \boxed{y = e^{x^2} + 2x e^{x^2}}$$

21.

$$\underbrace{x^2 y'' - 3xy' + ky = 0}_{\text{Cauchy-Euler DE}}$$

$$a = 1$$

$$b = -3$$

$$c = k$$

$$a m^2 + (b - a)m + c = 0$$

$$m^2 - 4m + k = 0$$

$$(a) \quad k = -5 \rightarrow m^2 - 4m - 5 = 0$$

$$(m - 5)(m + 1) = 0$$

$$m = 5, -1$$

$$y = c_1 x^5 + c_2 x^{-1}$$

$$(b) \quad k = 4 \rightarrow m^2 - 4m + 4 = 0$$

$$(m - 2)^2 = 0$$

$$m = 2 \quad [\text{repeated}]$$

$$y = c_1 x^2 + c_2 x^2 \ln x$$

$$(c) \quad k = 13 \rightarrow m^2 - 4m + 13 = 0$$

$$m^2 - 4m + 13 = -9$$

$$(m - 2)^2 = -9$$

$$m = 2 \pm 3i$$

$$y = x^2 \left[c_1 \cos(3 \ln x) + c_2 \sin(3 \ln x) \right]$$

$$22. (a) \quad y'' + 4y = 5x + \sin(2x)$$

$$m^2 + 4 = 0$$

$$m = \pm 2i$$

$$y_c = c_1 \cos(2x) + c_2 \sin(2x)$$

$$\therefore y_p = \underbrace{Ax + B}_{\text{for } 5x} + \underbrace{Cx \cos(2x) + Dx \sin(2x)}_{\text{Bad case for } \sin(2x)}$$

$$(b) \quad y'' - 2y' + y = 3x^2 + 5xe^x$$

$$m^2 - 2m + 1 = 0$$

$$(m-1)^2 = 0$$

$$y_c = c_1 e^x + c_2 x e^x$$

$$\therefore y_p = \underbrace{Ax^2 + Bx + C}_{\text{for } 3x^2} + \underbrace{x^2 [Dx + E] e^x}_{\text{Bad case for } 5xe^x}$$

$$(c) \quad y'' + 2y' + 5y = 3e^{-x} + 4e^{-x} \sin(2x) - 2\cos(3x)$$

$$m^2 + 2m + 5 = 0$$

$$m = -1 \pm 2i$$

$$y_c = e^{-x} \left(c_1 \cos(2x) + c_2 \sin(2x) \right)$$

$$\therefore y_p = \underbrace{Ae^{-x}}_{\text{for } 3e^{-x}} + \underbrace{xe^{-x} [B\cos(2x) + C\sin(2x)]}_{\text{Bad Case for } 4e^{-x} \sin(2x)} + \underbrace{[D\cos(3x) + E\sin(3x)]}_{\text{for } -2\cos(3x)}$$

Always find y_c to detect potential bad cases.

$$23. \quad y'' + y' - 6y = 4e^{3x} + 5\cos(2x)$$

step 1 Find y_c

$$m^2 + m - 6 = 0$$

$$(m+3)(m-2) = 0$$

$$m = -3, 2$$

$$y_c = c_1 e^{-3x} + c_2 e^{2x}$$

step 2 Find y_p

$$y_p = A e^{3x} + B \cos(2x) + C \sin(2x) \quad [\text{No bad case}]$$

$$y_p' = 3A e^{3x} - 2B \sin(2x) + 2C \cos(2x)$$

$$y_p'' = 9A e^{3x} - 4B \cos(2x) - 4C \sin(2x)$$

$$y_p'' + y_p' - 6y_p = 4e^{3x} + 5\cos(2x) + 0\sin(2x)$$

$$(9A + 3A - 6A)e^{3x} + [-4B + 2C - 6B]\cos(2x) + [-4C - 2B - 6C]\sin(2x)$$

$$\Rightarrow \begin{cases} 6A = 4 \\ -10B + 2C = 5 \\ -2B - 10C = 0 \end{cases} \rightarrow \begin{aligned} A &= 2/3 \\ B &= -25/52 \\ C &= 5/52 \end{aligned}$$

$$y = y_c + y_p$$

$$y = c_1 e^{-3x} + c_2 e^{2x} + \frac{2}{3} e^{3x} - \frac{25}{52} \cos(2x) + \frac{5}{52} \sin(2x)$$

24. $y'' - 5y' + 6y = x + 2 + 3e^{2x}$

Step 1 Find y_c

$$m^2 - 5m + 6 = 0$$

$$(m-3)(m-2) = 0$$

$$m = 3, 2$$

$$y_c = c_1 e^{3x} + c_2 e^{2x}$$

Step 2 Find y_p

$$y_p = \underbrace{Ax + B}_{\text{For } x+2} + \underbrace{Cxe^{2x}}_{\text{Bad Case for } 3e^{2x}}$$

$$y_p' = A + Ce^{2x} + 2Cxe^{2x}$$

$$y_p'' = 2Ce^{2x} + 2Ce^{2x} + 4Cxe^{2x}$$

$$y_p'' = 4Ce^{2x} + 4Cxe^{2x}$$

Substitute y_p, y_p', y_p'' into the DE to get:

$$\left[4Ce^{2x} + \cancel{4Cxe^{2x}} \right] - 5 \left[A + Ce^{2x} + \cancel{2Cxe^{2x}} \right] + 6 \left[Ax + B + \cancel{Cxe^{2x}} \right] = x + 2 + 3e^{2x}$$

$$4C - 5C = 3 \rightarrow C = -3$$

$$-5A + 6B = 2 \rightarrow B = \frac{2+5A}{6} = \frac{17}{36}$$

$$6A = 1 \rightarrow A = 1/6$$

$$\Rightarrow y = c_1 e^{3x} + c_2 e^{2x} + \frac{x}{6} + \frac{17}{36} - 3xe^{2x}$$

$$25. \quad y'' + 4y = \sec(2x)$$

step 1 Find y_c

$$m^2 + 4 = 0$$

$$m = \pm 2i$$

$$y_c = C_1 \cos(2x) + C_2 \sin(2x)$$

step 2 Find y_p with Variation of parameters

We look for y_p as $y_p = u_1 y_1 + u_2 y_2$

$$\begin{cases} y_1 = \cos(2x) \\ y_2 = \sin(2x) \\ f(x) = \sec(2x) \end{cases}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos(2x) & \sin(2x) \\ -2\sin(2x) & 2\cos(2x) \end{vmatrix}$$

$$W = 2\cos^2(2x) + 2\sin^2(2x) = 2$$

$$W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix} = \begin{vmatrix} 0 & \sin(2x) \\ \sec(2x) & 2\cos(2x) \end{vmatrix} = -\tan(2x)$$

$$W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix} = \begin{vmatrix} \cos(2x) & 0 \\ -2\sin(2x) & \sec(2x) \end{vmatrix} = 1$$

$$u_1 = \int \frac{W_1}{W} dx = \int \frac{\tan(2x)}{2} dx = \frac{1}{4} \ln |\cos(2x)|$$

$$u_2 = \int \frac{W_2}{W} dx = \int \frac{1}{2} dx = \frac{1}{2} x$$

$$\therefore \boxed{y = C_1 \cos(2x) + C_2 \sin(2x) + \frac{1}{2} x \sin(2x) + \frac{1}{4} \cos(2x) \ln |\cos(2x)|}$$

$$26. \quad x^2 y'' - 2xy' + 2y = x^4 e^{-x}$$

$$y'' - \frac{2}{x} y' + \frac{2}{x^2} y = x^2 e^{-x} \quad [\text{std form}]$$

step 1 Find y_c

$$\text{Cauchy-Euler: } m^2 - 3m + 2 = 0$$

$$(m-2)(m-1) = 0$$

$$m = 2, 1$$

$$y_c = c_1 x^2 + c_2 x$$

step 2 Find y_p

$$\begin{cases} y_1 = x^2 \\ y_2 = x \\ f(x) = x^2 e^{-x} \end{cases} \quad W = \begin{vmatrix} x^2 & x \\ 2x & 1 \end{vmatrix} = -x^2$$

$$W_1 = \begin{vmatrix} 0 & x \\ x^2 e^{-x} & 1 \end{vmatrix} = -x^3 e^{-x}$$

$$W_2 = \begin{vmatrix} x^2 & 0 \\ 2x & x^2 e^{-x} \end{vmatrix} = x^4 e^{-x}$$

$$u_1 = \int \frac{W_1}{W} dx = \int x e^{-x} dx = -(x+1) e^{-x}$$

$$u_2 = \int \frac{W_2}{W} dx = \int -x^2 e^{-x} dx = (x^2 + 2x + 2) e^{-x}$$

$$y_p = u_1 y_1 + u_2 y_2 = -(x+1)x^2 e^{-x} + (x^2 + 2x + 2)x e^{-x}$$

$$y_p = (x^2 + 2x) e^{-x}$$

$$\therefore \boxed{y = c_1 x^2 + c_2 x + (x^2 + 2x) e^{-x}}$$

$$27. \quad x^2 y'' - x y' + y = 2x, \quad y(1) = 2, \quad y'(1) = 3$$

Step 1 Find y_c

$$\begin{aligned} \text{Cauchy-Euler: } m^2 - 2m + 1 &= 0 \\ (m-1)^2 &= 0 \\ m &= 1 \end{aligned}$$

$$y_c = c_1 x + c_2 x \ln x$$

Step 2 Find y_p

$$\begin{cases} y_1 = x \\ y_2 = x \ln x \\ f(x) = 2/x \quad [\text{From standard form}] \end{cases}$$

$$W = \begin{vmatrix} x & x \ln x \\ 1 & \ln x + 1 \end{vmatrix}$$

$$W = x$$

$$W_1 = \begin{vmatrix} 0 & x \ln x \\ 2/x & \ln x + 1 \end{vmatrix} = -2 \ln x$$

$$W_2 = \begin{vmatrix} x & 0 \\ 1 & 2/x \end{vmatrix} = 2$$

$$u_1 = \int \frac{W_1}{W} dx = \int \frac{-2 \ln x}{x} dx = -(\ln x)^2$$

$$u_2 = \int \frac{W_2}{W} dx = \int \frac{2}{x} dx = 2 \ln x \quad (\text{since } x > 0)$$

$$y_p = u_1 y_1 + u_2 y_2 = -x (\ln x)^2 + 2x (\ln x) = x (\ln x)^2$$

Step 3 Find c_1, c_2

$$y = c_1 x + c_2 x \ln x + x (\ln x)^2$$

$$y' = c_1 + c_2 (\ln x + 1) + (\ln x)^2 + 2 \ln x$$

$$y(1) = 2 \rightarrow c_1 = 2$$

$$y'(1) = 3 \rightarrow c_1 + c_2 = 3 \rightarrow c_2 = 1$$

$$\therefore y = 2x + x \ln x + x (\ln x)^2$$

28. $m x''(t) + \beta x'(t) + k x(t) = F(t)$

Here: $m = 3$, $\underbrace{\beta = 0}_{\text{No damping}}$, $k = 12$, $\underbrace{F(t) = 0}_{\text{No external force}}$

$$3x''(t) + 12x(t) = 0$$

$$x''(t) + 4x(t) = 0$$

$$\rightarrow x(t) = c_1 \cos(2t) + c_2 \sin(2t) \quad [\text{in meters}]$$

$$x(0) = +20 \text{ cm} = +0.2 \text{ m} \quad [+ \text{ since below}]$$

$$x'(0) = -0.6 \text{ m/s} \quad [- \text{ since upward}]$$

$$x'(t) = -2c_1 \sin(2t) + 2c_2 \cos(2t)$$

$$x(0) = 0.2 \rightarrow c_1 = 0.2$$

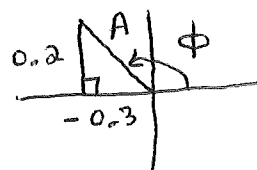
$$x'(0) = -0.6 \rightarrow 2c_2 = -0.6 \rightarrow c_2 = -0.3$$

$$(a) \quad x(t) = 0.2 \cos(2t) - 0.3 \sin(2t)$$

$$(b) \quad \text{Period: } T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi \text{ seconds}$$

$$(c) \quad x(t) = A \sin(\omega t + \phi) = A \sin \phi \cos(\omega t) + A \cos \phi \sin(\omega t)$$

$$\omega = 2 \quad \text{and} \quad \begin{cases} A \sin \phi = 0.2 \\ A \cos \phi = -0.3 \end{cases}$$



$$A = \sqrt{0.2^2 + 0.3^2}, \quad \phi = \text{Arctan}\left(\frac{0.2}{-0.3}\right) + \pi = 2.55$$

$$\text{The lowest point is when } x(t) = A \rightarrow \omega t + \phi = \frac{\pi}{2} + 2n\pi$$

$$\text{The first } t > 0 \text{ is: } t = \frac{\frac{\pi}{2} + 2\pi - \phi}{\omega} = \boxed{2.65 \text{ seconds}}$$

$$29. \quad m x''(t) + \beta x'(t) + K x(t) = F(t)$$

$$2 x''(t) + 5 x'(t) + 3 x(t) = 4 \cos(2t) + 3 \sin(2t)$$

since there is damping ($\beta > 0$), $\lim_{t \rightarrow \infty} x_c(t) = 0$

$$x(t) = \underbrace{x_c(t)}_{\text{Transient}} + \underbrace{x_p(t)}_{\text{steady-state}}$$

$$(a) \quad x_p = a \cos(2t) + b \sin(2t)$$

$$x'_p = -2a \sin(2t) + 2b \cos(2t)$$

$$x''_p = -4a \cos(2t) - 4b \sin(2t)$$

substitute in the DE and collect terms to get:

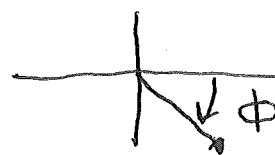
$$(-8a + 10b + 3a) \cos(2t) + (-8b - 10a + 3b) \sin(2t) = 4 \cos(2t) + 3 \sin(2t)$$

$$\Rightarrow \begin{cases} -5a + 10b = 4 \\ -10a - 5b = 3 \end{cases} \xrightarrow{\text{solve}} \begin{cases} a = -2/5 \\ b = 1/5 \end{cases}$$

$$\text{steady-state : } \boxed{x_p = -\frac{2}{5} \cos(2t) + \frac{1}{5} \sin(2t)}$$

$$(b) \quad A \sin(2t + \phi) = A \sin \phi \cos(2t) + A \cos \phi \sin(2t)$$

$$\Rightarrow \begin{cases} A \sin \phi = -2/5 \\ A \cos \phi = 1/5 \end{cases}$$



$$\phi = \arctan\left(\frac{-2/5}{1/5}\right)$$

$$\phi = -1.107$$

$$A = \sqrt{\left(-\frac{2}{5}\right)^2 + \left(\frac{1}{5}\right)^2} = \frac{\sqrt{5}}{5}$$

$$\therefore \boxed{x_p = \frac{\sqrt{5}}{5} \sin(2t - 1.107)}$$

$$30. \quad m x''(t) + \beta x'(t) + k x(t) = F(t)$$

$$m = 1 \text{ kg}, \quad k = 2 \text{ N/m}, \quad \beta = 3 \text{ N/(m/s)}$$

$$(a) \quad F(t) = 0. \quad x'' + 3x' + 2x = 0$$

$$M^2 + 3M + 2 = 0$$

$$(M+2)(M+1) = 0$$

$$M = -2, -1$$

$$\Rightarrow x(t) = c_1 e^{-2t} + c_2 e^{-t}$$

$$x'(t) = -2c_1 e^{-2t} - c_2 e^{-t}$$

$$x(0) = +50 \text{ cm} = 0.5 \text{ m}$$

$$x'(0) = 0 \quad [\text{No initial velocity}]$$

$$\Rightarrow \begin{cases} c_1 + c_2 = 0.5 \\ -2c_1 - c_2 = 0 \end{cases}$$

$$\text{Solve to get: } \begin{cases} c_1 = -0.5 \\ c_2 = 1 \end{cases}$$

$$\boxed{x(t) = -0.5 e^{-2t} + e^{-t}}$$

$$(b) \quad x_p(t) \text{ is the steady-state since } \lim_{t \rightarrow \infty} x_c = 0$$

$$\begin{cases} x_p = a \cos t + b \sin t \\ x'_p = -a \sin t + b \cos t \\ x''_p = -a \cos t - b \sin t \end{cases}$$

$$\boxed{F(t) = F_0 \cos t}$$

$$x'' + 3x' + 2x = F_0 \cos t$$

$$\text{plug in the DE to get: } \begin{cases} -a + 3b + 2a = F_0 \rightarrow a + 3b = F_0 \\ -b - 3a + 2b = 0 \rightarrow -3a + b = 0 \end{cases}$$

$$\Rightarrow \begin{cases} a = F_0/10 \\ b = 3F_0/10 \end{cases}$$

$$\text{Amplitude} = \sqrt{a^2 + b^2} = \sqrt{\left(\frac{F_0}{10}\right)^2 + \left(\frac{3F_0}{10}\right)^2}$$

$$= \frac{F_0 \sqrt{10}}{10}$$

$$\text{Amplitude} = 1 \rightarrow \boxed{F_0 = \sqrt{10}}$$

31. $m = 1 \text{ kg}$, $\beta = 4 \text{ N/(m/s)}$, $k = 5 \text{ N/m}$, $\underbrace{F = 0}_{\text{No ext. force}}$

$$x'' + 4x' + 5x = 0$$

$$m^2 + 4m + 5 = 0$$

$$m^2 + 4m + 4 = -1$$

$$(m+2)^2 = -1$$

$$m = -2 \pm i$$

$$x(t) = e^{-2t} (c_1 \cos t + c_2 \sin t)$$

$$x'(t) = -2e^{-2t} (c_1 \cos t + c_2 \sin t) + e^{-2t} (-c_1 \sin t + c_2 \cos t)$$

(a) $x(0) = 50 \text{ cm} = 0.5 \text{ m}$

$$x'(0) = 1 \text{ m/s}$$

$$x(0) = 0.5 \rightarrow c_1 = 0.5$$

$$x'(0) = 1 \rightarrow -2c_1 + c_2 = 1 \rightarrow c_2 = 1 + 2c_1 = 2$$

$$x(t) = e^{-2t} (0.5 \cos t + 2 \sin t)$$

(b) $x(t) = A e^{-2t} \sin(t + \phi)$

$$\begin{cases} A \sin \phi = 0.5 \\ A \cos \phi = 2 \end{cases}$$



$$\phi = \text{Arctan}\left(\frac{0.5}{2}\right)$$

$$\phi = 0.245$$

$$\underbrace{x(t) = 0}_{\text{equilibrium}} \rightarrow \sin(t + \phi) = 0 \rightarrow t + \phi = n\pi$$

$$t = n\pi - \phi$$

the positive t are :

$$t = n\pi - 0.245$$

$$\text{for } n = 1, 2, 3, \dots$$

$$3a. \quad \begin{cases} x''(t) + 5x'(t) + 6x(t) = 3\sin(2t) - \cos(2t) \\ x(0) = -0.5 \text{ m} \\ x'(0) = 1 \text{ m/s} \end{cases}$$

Step 1 Find x_c

$$M^2 + 5M + 6 = 0$$

$$(M+2)(M+3) = 0$$

$$M = -2, -3$$

$$x_c = c_1 e^{-2t} + c_2 e^{-3t}$$

Step 2 Find x_p

$$x_p = A \sin(2t) + B \cos(2t)$$

$$x'_p = 2A \cos(2t) - 2B \sin(2t)$$

$$x''_p = -4A \sin(2t) - 4B \cos(2t)$$

Substitute in the DE and collect terms to get:

$$(2A - 10B) \sin(2t) + (10A - 2B) \cos(2t) = 3\sin(2t) - \cos(2t)$$

$$\Rightarrow \begin{cases} 2A - 10B = 3 \\ 10A - 2B = -1 \end{cases} \xrightarrow{\text{solve}} \begin{cases} A = -1/26 \\ B = -4/13 \end{cases}$$

$$x = \underbrace{c_1 e^{-2t} + c_2 e^{-3t}}_{x_c} - \underbrace{\frac{1}{26} \sin(2t) - \frac{4}{13} \cos(2t)}_{x_p}$$

Step 3 Find c_1, c_2

$$x' = -2c_1 e^{-2t} - 3c_2 e^{-3t} - \frac{1}{13} \cos(2t) + \frac{8}{13} \sin(2t)$$

$$\left. \begin{aligned} x(0) = -0.5 &\rightarrow c_1 + c_2 - \frac{4}{13} = -\frac{1}{2} \rightarrow c_1 + c_2 = -5/26 \\ x'(0) = 1 &\rightarrow -2c_1 - 3c_2 - \frac{1}{13} = 1 \rightarrow 2c_1 + 3c_2 = -14/13 \end{aligned} \right\}$$

Solve to get: $c_1 = 1/2$ and $c_2 = -9/13$

$$\therefore x = \frac{1}{2} e^{-2t} - \frac{9}{13} e^{-3t} - \frac{4}{13} \cos(2t) - \frac{1}{26} \sin(2t)$$

33. $y'' + x^2 y' - 3y = 0$, $y(0) = 3$, $y'(0) = -2$

Let $y = \sum_{n=0}^{\infty} a_n x^n \rightarrow y' = \sum_{n=1}^{\infty} n a_n x^{n-1} \rightarrow y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$

$$\Rightarrow \underbrace{\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}}_{\substack{k=n-2 \\ n=k+2}} + \underbrace{\sum_{n=1}^{\infty} n a_n x^{n+1}}_{\substack{k=n+1 \\ n=k-1}} - \underbrace{\sum_{n=0}^{\infty} 3 a_n x^n}_{k=n} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} x^k + \sum_{k=2}^{\infty} (k-1) a_{k-1} x^k - \sum_{k=0}^{\infty} 3 a_k x^k = 0$$

$$\Rightarrow (2a_2 - 3a_0) + (6a_3 - 3a_1)x + \sum_{k=2}^{\infty} [(k+2)(k+1)a_{k+2} + (k-1)a_{k-1} - 3a_k] x^k = 0$$

$$\Rightarrow \begin{cases} 2a_2 - 3a_0 = 0 \\ 6a_3 - 3a_1 = 0 \\ (k+2)(k+1)a_{k+2} + (k-1)a_{k-1} - 3a_k = 0 \text{ for } k=2, 3, 4, \dots \end{cases}$$

$$\Rightarrow \begin{cases} a_2 = 3a_0/2 \\ a_3 = a_1/2 \\ a_{k+2} = \frac{3a_k - (k-1)a_{k-1}}{(k+2)(k+1)} \text{ for } k=2, 3, \dots \end{cases}$$

From the initial conditions we have: $a_0 = 3$, $a_1 = -2$

$$\Rightarrow a_2 = 9/2, \quad a_3 = -1$$

$$k=2 \rightarrow a_4 = \frac{3a_2 - a_1}{4(3)} = 31/24$$

$$\therefore y = 3 - 2x + \frac{9}{2}x^2 - x^3 + \frac{31}{24}x^4 + \dots$$

The DE has no S.P., so the series converges for all x .

$$34. \quad (x+1)y'' - (2-x)y' + y = 0, \quad y(0)=2, \quad y'(0)=-1$$

$$y = \sum_{n=0}^{\infty} a_n x^n \rightarrow y' = \sum_{n=1}^{\infty} n a_n x^{n-1} \rightarrow y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

$$x y'' + y'' - 2y' + x y' + y = 0$$

$$\underbrace{\sum_{n=2}^{\infty} n(n-1) a_n x^{n-1}}_{\substack{k=n-1 \\ n=k+1}} + \underbrace{\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}}_{\substack{k=n-2 \\ n=k+2}} - \underbrace{\sum_{n=1}^{\infty} 2n a_n x^{n-1}}_{\substack{k=n-1 \\ n=k+1}} + \underbrace{\sum_{n=1}^{\infty} n a_n x^n}_{k=n} + \underbrace{\sum_{n=0}^{\infty} a_n x^n}_{k=n} = 0$$

$$\sum_{k=1}^{\infty} (k+1)k a_{k+1} x^k + \sum_{k=0}^{\infty} (k+2)(k+1) a_{k+2} x^k - \sum_{k=0}^{\infty} 2(k+1) a_{k+1} x^k + \sum_{k=1}^{\infty} k a_k x^k + \sum_{k=0}^{\infty} a_k x^k = 0$$

$$(2a_2 - 2a_1 + a_0) + \sum_{k=1}^{\infty} [(k+1)k a_{k+1} + (k+2)(k+1) a_{k+2} - 2(k+1) a_{k+1} + k a_k + a_k] x^k = 0$$

$$\Rightarrow 2a_2 - 2a_1 + a_0 = 0$$

$$(k+1)k a_{k+1} + (k+2)(k+1) a_{k+2} - 2(k+1) a_{k+1} + (k+1) a_k = 0$$

$$(k+2)(\cancel{k+1}) a_{k+2} + (k-2)(\cancel{k+1}) a_{k+1} + (\cancel{k+1}) a_k = 0$$

$$\Rightarrow \begin{cases} a_2 = \frac{2a_1 - a_0}{2} \\ a_{k+2} = \frac{(2-k) a_{k+1} - a_k}{(k+2)} \quad \text{for } k=1, 2, \dots \end{cases}$$

$$y(0) = 2 \rightarrow a_0 = 2$$

$$y'(0) = -1 \rightarrow a_1 = -1$$

$$a_2 = \frac{2a_1 - a_0}{2} = -2$$

$$(k=1) \quad a_3 = \frac{a_2 - a_1}{3} = -1/3$$

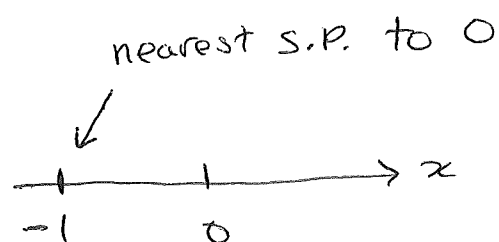
$$(k=2) \quad a_4 = \frac{0 - a_2}{4} = 1/2$$

$$(k=3) \quad a_5 = \frac{-a_4 - a_3}{5} = -1/30$$

$$\therefore \boxed{y = 2 - x - 2x^2 - \frac{x^3}{3} + \frac{x^4}{2} - \frac{x^5}{30} + \dots}$$

$$y'' - \frac{(2-x)}{(x+1)} y' + \frac{1}{(x+1)} y = 0$$

$x = -1$ is the only S.P.



The series converges at least for

$$\boxed{-1 < x < 1}$$

$$35. \quad y'' - (1+x)y = 0 \rightarrow y'' - y - xy = 0$$

$$\underbrace{\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}}_{\substack{k=n-2 \\ n=k+2}} - \underbrace{\sum_{n=0}^{\infty} a_n x^n}_{k=n} - \underbrace{\sum_{n=0}^{\infty} a_n x^{n+1}}_{\substack{k=n+1 \\ n=k-1}} = 0$$

$$\sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2} x^k - \sum_{k=0}^{\infty} a_k x^k - \sum_{k=1}^{\infty} a_{k-1} x^k = 0$$

$$(2a_2 - a_0) + \sum_{k=1}^{\infty} [(k+2)(k+1)a_{k+2} - a_k - a_{k-1}] x^k = 0$$

$$\Rightarrow \begin{cases} 2a_2 - a_0 = 0 \\ (k+2)(k+1)a_{k+2} - a_k - a_{k-1} = 0 \quad \text{for } k=1, 2, \dots \end{cases}$$

$$\Rightarrow \begin{cases} a_2 = a_0/2 \\ a_{k+2} = \frac{a_k + a_{k-1}}{(k+2)(k+1)} \quad k=1, 2, 3, \dots \end{cases}$$

Let's now find two LI

Series Solutions: y_1 and y_2 .

y_1	y_2
$a_0 = 1$	$a_0 = 0$
$a_1 = 0$	$a_1 = 1$
$a_2 = \frac{a_0}{2} = \frac{1}{2}$	$a_2 = 0$
$a_3 = \frac{a_1 + a_0}{6} = \frac{1}{6}$	$a_3 = \frac{1}{6}$
$a_4 = \frac{a_2 + a_1}{12} = \frac{1}{24}$	$a_4 = \frac{1}{12}$
$\vdots$	$\vdots$

$$y = c_1 y_1 + c_2 y_2$$

$$y = c_1 \left[1 + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots \right] + c_2 \left[x + \frac{x^3}{6} + \frac{x^4}{12} + \dots \right]$$

Both series converges for all x since the DE has no S.P.

$$36. \quad y'' + e^x y = 0$$

$$\text{Let } y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$$

$$\text{then } \begin{cases} y' = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \dots \\ y'' = 2a_2 + 6a_3 x + 12a_4 x^2 + \dots \end{cases}$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$

$$y'' + e^x y = 0$$

$$\left[2a_2 + 6a_3 x + 12a_4 x^2 + \dots \right] + \left[1 + x + \frac{x^2}{2} + \dots \right] \left[a_0 + a_1 x + a_2 x^2 + \dots \right]$$

$$\dots = 0$$

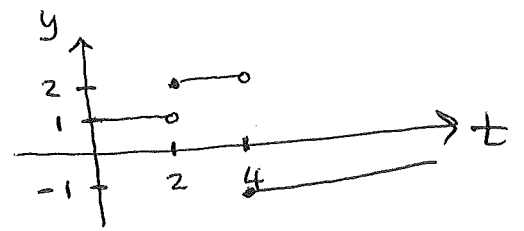
$$\Rightarrow \begin{cases} 2a_2 + a_0 = 0 \\ 6a_3 + a_1 + a_0 = 0 \\ 12a_4 + a_2 + a_1 + \frac{a_0}{2} = 0 \end{cases} \Rightarrow \begin{cases} a_2 = -a_0/2 \\ a_3 = -(a_1 + a_0)/6 \\ a_4 = -\frac{(a_2 + a_1 + a_0/2)}{12} \end{cases}$$

y_1	y_2
$a_0 = 1$	$a_0 = 0$
$a_1 = 0$	$a_1 = 1$
$a_2 = -1/2$	$a_2 = 0$
$a_3 = -1/6$	$a_3 = -1/6$
$a_4 = 0$	$a_4 = -1/12$

$$y = C_1 y_1 + C_2 y_2$$

$$\therefore y = C_1 \left[1 - \frac{x^2}{2} - \frac{x^3}{6} + \dots \right] + C_2 \left[x - \frac{x^3}{6} - \frac{x^4}{12} + \dots \right]$$

$$37. (a) f(t) = \begin{cases} 1, & 0 \leq t < 2 \\ 2, & 2 \leq t < 4 \\ -1, & t \geq 4 \end{cases}$$



$$f(t) = 1 + u(t-2) - 3u(t-4)$$

$$\mathcal{L}\{f(t)\} = \frac{1}{s} + \frac{e^{-2s}}{s} - \frac{3e^{-4s}}{s}$$

$$(b) \mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\} = e^{-3t}$$

$$\Rightarrow \mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s+3}\right\} = e^{-3(t-2)} u(t-2)$$

$$(c) \mathcal{L}\{t \sin(3t)\} = \frac{6s}{(s^2+9)^2}$$

$$\Rightarrow \mathcal{L}\{t e^{-at} \sin(3t)\} = \frac{6(s+a)}{[(s+a)^2+9]^2}$$

$$(d) \frac{s+4}{s^2+2s+5} = \frac{s+4}{(s+1)^2+4} = \frac{(s+1)+3}{(s+1)^2+4}$$

$$= \frac{s+1}{(s+1)^2+4} + \frac{3}{2} \left[\frac{2}{(s+1)^2+4} \right]$$

$$\Rightarrow \mathcal{L}^{-1}\left\{\frac{s+4}{s^2+2s+5}\right\} = e^{-t} \cos(2t) + \frac{3}{2} e^{-t} \sin(2t)$$

$$(e) \mathcal{L}\{\cos t \, u(t-\pi)\} = e^{-as} \mathcal{L}\{\cos(t+\pi)\}$$

$$\boxed{a=\pi}$$

$$= e^{-as} \mathcal{L}\{-\cos t\}$$

$$= \boxed{\frac{-s e^{-\pi s}}{s^2+1}}$$

$$38. \quad \begin{cases} y''(t) + 4y'(t) + 3y(t) = 3e^{-2t} \\ y(0) = 2 \\ y'(0) = 3 \end{cases}$$

Take Laplace on both sides of the DE

$$\left[s^2 Y(s) - sy(0) - y'(0) \right] + 4 \left[sY(s) - y(0) \right] + 3Y(s) = 3 \left(\frac{1}{s+2} \right)$$

$$(s^2 + 4s + 3)Y(s) - 2s - 3 - 8 = \frac{3}{s+2}$$

$$Y(s) = \frac{2s+11}{s^2+4s+3} + \frac{3}{(s+2)(s^2+4s+3)}$$

$$Y(s) = \frac{(2s+11)(s+2) + 3}{(s+1)(s+3)(s+2)} = \frac{A}{s+1} + \frac{B}{s+3} + \frac{C}{s+2}$$

Cover-up

$$A = \frac{(-2+11)(1)+3}{2(1)} = 6 \quad \left[\text{set } s = -1 \right]$$

$$B = \frac{(-6+11)(-1)+3}{(-2)(-1)} = -1 \quad \left[\text{set } s = -3 \right]$$

$$C = \frac{0+3}{(-1)(1)} = -3 \quad \left[\text{set } s = -2 \right]$$

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = 6e^{-t} - e^{-3t} - 3e^{-2t}$$

39.

$$y'(t) + 3y(t) = \sin(2t), \quad y(0) = 4.$$

$$\rightarrow [sY(s) - y(0)] + 3Y(s) = \frac{2}{s^2+4}$$

$$\rightarrow (s+3)Y(s) - 4 = \frac{2}{s^2+4}$$

$$\rightarrow Y(s) = \frac{4}{s+3} + \boxed{\frac{2}{(s+3)(s^2+4)}} \quad \leftarrow \begin{array}{l} \text{requires} \\ \text{P.F.} \end{array}$$

$$\frac{2}{(s+3)(s^2+4)} = \frac{A}{s+3} + \frac{Bs+C}{s^2+4} \quad \xrightarrow{\text{Solve}} \begin{cases} A = 2/13 \\ B = -2/13 \\ C = 6/13 \end{cases}$$

$$Y(s) = \frac{4}{s+3} + \frac{2/13}{s+3} + \frac{(-2/13)s + 6/13}{s^2+4}$$

$$Y(s) = \frac{54}{13} \left(\frac{1}{s+3} \right) - \frac{2}{13} \left(\frac{s}{s^2+4} \right) + \frac{3}{13} \left(\frac{2}{s^2+4} \right)$$

$$y(t) = \mathcal{L}^{-1} \{ Y(s) \}$$

$$y(t) = \frac{54}{13} e^{-3t} - \frac{2}{13} \cos(2t) + \frac{3}{13} \sin(2t)$$

$$40. \begin{cases} y''(t) + 9y(t) = e^{-t} + \cos(3t) \\ y(0) = 0, \quad y'(0) = 1 \end{cases}$$

$$\left[s^2 Y(s) - sy(0) - y'(0) \right] + 9Y(s) = \frac{1}{s+1} + \frac{s}{s^2+9}$$

$$(s^2+9)Y(s) - 1 = \frac{1}{s+1} + \frac{s}{s^2+9}$$

$$Y(s) = \frac{1}{s^2+9} + \frac{1}{(s+1)(s^2+9)} + \frac{s}{(s^2+9)^2}$$

P.F. $\frac{1}{(s+1)(s^2+9)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+9} \rightarrow \begin{cases} A = 1/10 \\ B = -1/10 \\ C = 1/10 \end{cases}$

$$Y(s) = \frac{1}{s^2+9} + \frac{1/10}{s+1} + \frac{(-1/10)s + 1/10}{s^2+9} + \frac{s}{(s^2+9)^2}$$

$$\Rightarrow Y(s) = \frac{11}{30} \left[\frac{3}{s^2+9} \right] + \frac{1}{10} \left[\frac{1}{s+1} \right] - \frac{1}{10} \left[\frac{s}{s^2+9} \right] + \frac{1}{6} \left[\frac{6s}{(s^2+9)^2} \right]$$

$$y(t) = \mathcal{L}^{-1}\{Y(s)\}$$

$$y(t) = \frac{11}{30} \sin(3t) + \frac{1}{10} e^{-t} - \frac{1}{10} \cos(3t) + \frac{1}{6} t \sin(3t)$$

$$41. \begin{cases} y''(t) + 2y'(t) + 5y(t) = 3e^{-t} \cos(2t) \\ y(0) = 1, \quad y'(0) = 2 \end{cases}$$

$$[s^2 Y(s) - sy(0) - y'(0)] + 2[sY(s) - y(0)] + 5Y(s) = \frac{3(s+1)}{(s+1)^2 + 4}$$

$$(s^2 + 2s + 5)Y(s) - s - 2 - 2 = \frac{3(s+1)}{(s+1)^2 + 4}$$

Complete the square: $s^2 + 2s + 5 = (s+1)^2 + 4$

$$Y(s) = \frac{s+4}{(s+1)^2 + 4} + \frac{3(s+1)}{[(s+1)^2 + 4]^2}$$

$$Y(s) = \frac{(s+1) + 3}{(s+1)^2 + 4} + \frac{3(s+1)}{[(s+1)^2 + 4]^2}$$

$$Y(s) = \frac{s+1}{(s+1)^2 + 4} + \frac{3}{2} \left[\frac{2}{(s+1)^2 + 4} \right] + \frac{3}{4} \left[\frac{4(s+1)}{[(s+1)^2 + 4]^2} \right]$$

$$y(t) = \mathcal{L}^{-1}\{Y(s)\}$$

$$y(t) = e^{-t} \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 4} + \frac{3}{2} \left(\frac{2}{s^2 + 4} \right) + \frac{3}{4} \left[\frac{4s}{(s^2 + 4)^2} \right] \right\}$$

$$y(t) = e^{-t} \cos(2t) + \frac{3}{2} e^{-t} \sin(2t) + \frac{3}{4} t e^{-t} \sin(2t)$$

$$42. \begin{cases} y'(t) + 5y(t) = 3 + \delta(t-2) \\ y(0) = 4 \end{cases}$$

$$[sY(s) - y(0)] + 5Y(s) = \frac{3}{s} + e^{-2s}$$

$$(s+5)Y(s) - 4 = \frac{3}{s} + e^{-2s}$$

$$Y(s) = \frac{4}{s+5} + \left\{ \frac{3}{s(s+5)} \right\} + e^{-2s} \left(\frac{1}{s+5} \right)$$

$$\downarrow$$

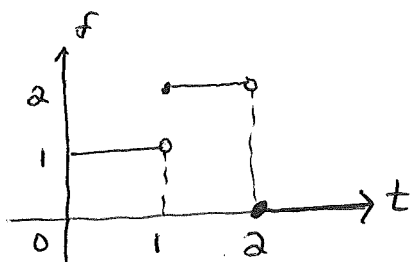
$$\frac{A}{s} + \frac{B}{s+5} \Rightarrow \begin{cases} A = 3/5 \\ B = -3/5 \end{cases}$$

$$Y(s) = \frac{4}{s+5} + \frac{3/5}{s} + \frac{(-3/5)}{s+5} + e^{-2s} \left(\frac{1}{s+5} \right)$$

$$Y(s) = \frac{17}{5} \left(\frac{1}{s+5} \right) + \frac{3}{5} \left(\frac{1}{s} \right) + e^{-2s} \left(\frac{1}{s+5} \right)$$

$$y(t) = \frac{17}{5} e^{-5t} + \frac{3}{5} + e^{-5(t-2)} \mathcal{U}(t-2)$$

43. $y'(t) + 2y(t) = f(t)$, $y(0) = 0$



$$f(t) = 1 + u(t-1) - 2u(t-2)$$

$$[sY(s) - y(0)] + 2Y(s) = \frac{1}{s} + \frac{e^{-s}}{s} - \frac{2e^{-2s}}{s}$$

$$Y(s) = \frac{1}{s(s+2)} + e^{-s} \left(\frac{1}{s(s+2)} \right) - 2e^{-2s} \left(\frac{1}{s(s+2)} \right)$$

$$Y(s) = W(s) + e^{-s} W(s) - 2e^{-2s} W(s)$$

Where : $W(s) = \frac{1}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2} \quad \begin{cases} A = 1/2 \\ B = -1/2 \end{cases}$

$$y(t) = w(t) + w(t-1)u(t-1) - 2w(t-2)u(t-2)$$

Where : $w(t) = \frac{1}{2} - \frac{1}{2}e^{-2t}$

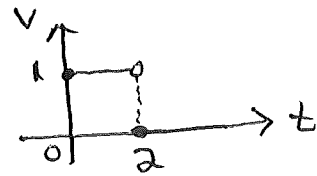
$$y(t) = \left(\frac{1}{2} - \frac{1}{2}e^{-2t} \right) + \left(\frac{1}{2} - \frac{1}{2}e^{-2(t-1)} \right) u(t-1) - \left(1 - e^{-2(t-2)} \right) u(t-2)$$

$$y(1.5) = \left(\frac{1}{2} - \frac{1}{2}e^{-3} \right) + \left(\frac{1}{2} - \frac{1}{2}e^{-1} \right)$$

$$\approx \boxed{0.791}$$

44.

$$V(t) = 1 - u(t-2)$$



$$\begin{cases} \frac{di}{dt} + 10i(t) = V(t) \\ i(0) = 0 \end{cases}$$

$$[sI(s) - i(0)] + 10I(s) = \frac{1}{s} - \frac{e^{-2s}}{s}$$

$$(s+10)I(s) = \frac{1}{s} - \frac{e^{-2s}}{s}$$

$$I(s) = \frac{1}{s(s+10)} - e^{-2s} \left(\frac{1}{s(s+10)} \right)$$

$$I(s) = W(s) - e^{-2s} W(s)$$

$$W(s) = \frac{1}{s(s+10)} = \frac{A}{s} + \frac{B}{s+10} \rightarrow \begin{cases} A = 1/10 \\ B = -1/10 \end{cases}$$

$$w(t) = \mathcal{L}^{-1}\{W(s)\} = \frac{1}{10} - \frac{1}{10} e^{-10t}$$

$$i(t) = \mathcal{L}^{-1}\{I(s)\} = w(t) - w(t-2)u(t-2)$$

$$i(t) = \left(\frac{1}{10} - \frac{1}{10} e^{-10t} \right) - \left(\frac{1}{10} - e^{-10(t-2)} \right) u(t-2)$$

$$\text{or } i(t) = \begin{cases} \frac{1}{10} - \frac{1}{10} e^{-10t}, & 0 \leq t < 2 \\ \frac{1}{10} (e^{-10(t-2)} - e^{-10t}), & t \geq 2 \end{cases}$$

$$45. \begin{cases} y''(t) + y(t) = 2e^{-t} + \delta(t - \pi/2) + \delta(t - \pi) \\ y(0) = 2 \\ y'(0) = 1 \end{cases}$$

$$[s^2 Y(s) - sy(0) - y'(0)] + Y(s) = \frac{2}{s+1} + e^{-\frac{\pi}{2}s} + e^{-\pi s}$$

$$(s^2 + 1)Y(s) - 2s - 1 = \frac{2}{s+1} + e^{-\frac{\pi}{2}s} + e^{-\pi s}$$

$$Y(s) = \frac{2s+1}{s^2+1} + \left\{ \frac{2}{(s+1)(s^2+1)} \right\} + \frac{e^{-\frac{\pi}{2}s}}{s^2+1} + \frac{e^{-\pi s}}{s^2+1}$$

$$\frac{A}{s+1} + \frac{Bs+C}{s^2+1} \rightarrow \begin{cases} A = 1 \\ B = -1 \\ C = 1 \end{cases}$$

$$Y(s) = \frac{2s+1}{s^2+1} + \frac{1}{s+1} + \frac{(-s+1)}{s^2+1} + \frac{e^{-\frac{\pi}{2}s}}{s^2+1} + \frac{e^{-\pi s}}{s^2+1}$$

$$Y(s) = \frac{1}{s+1} + \frac{s}{s^2+1} + \frac{2}{s^2+1} + \frac{e^{-\frac{\pi}{2}s}}{s^2+1} + \frac{e^{-\pi s}}{s^2+1}$$

$$y(t) = e^{-t} + \cos t + 2 \sin t + \sin(t - \frac{\pi}{2})u(t - \frac{\pi}{2}) \\ \dots + \sin(t - \pi)u(t - \pi)$$

$$y(t) = e^{-t} + \cos t + 2 \sin t - \cos t u(t - \frac{\pi}{2}) - \sin t u(t - \pi)$$

$$y(t) = \begin{cases} e^{-t} + \cos t + 2 \sin t & , \quad 0 \leq t < \pi/2 \\ e^{-t} + 2 \sin t & , \quad \pi/2 \leq t < \pi \\ e^{-t} + \sin t & , \quad t \geq \pi \end{cases}$$

$$46. \quad \mathcal{L}^{-1}\{F(s)G(s)\} = f(t) * g(t)$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} = e^{+3t}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s+4}\right\} = e^{-4t}$$

$$\Rightarrow \mathcal{L}^{-1}\left\{\frac{1}{(s-3)(s+4)}\right\} = e^{3t} * e^{-4t}$$

$$= \int_0^t e^{3\theta} e^{-4(t-\theta)} d\theta$$

$$= \int_0^t e^{7\theta} e^{-4t} d\theta$$

$$= e^{-4t} \left[\frac{e^{7\theta}}{7} \right]_0^t$$

$$= \frac{e^{-4t}}{7} [e^{7t} - 1]$$

$$= \boxed{\frac{1}{7} e^{3t} - \frac{1}{7} e^{-4t}}$$

$$47. \quad \mathcal{L}^{-1} \left\{ \frac{1}{s^2+1} \right\} = \sin t$$

$$\Rightarrow \mathcal{L}^{-1} \left\{ \frac{1}{(s^2+1)^2} \right\} = \sin t * \sin t$$

$$= \int_0^t \sin \theta \sin(t-\theta) d\theta$$

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$\begin{aligned} \Rightarrow \mathcal{L}^{-1} \left\{ \frac{1}{(s^2+1)^2} \right\} &= \int_0^t \frac{1}{2} [\cos(2\theta-t) - \cos t] d\theta \\ &= \frac{1}{2} \left[\frac{\sin(2\theta-t)}{2} - \theta \cos t \right] \Big|_0^t \\ &= \frac{1}{2} \left[\frac{\sin t}{2} - t \cos t \right] - \frac{1}{2} \left[\frac{\sin(-t)}{2} - 0 \right] \\ &= \frac{\sin t}{4} - \frac{t \cos t}{2} - \frac{1}{4} \sin(-t) \\ &= \frac{\sin t}{4} - \frac{t \cos t}{2} + \frac{1}{4} \sin t \\ &= \boxed{\frac{\sin t - t \cos t}{2}} \end{aligned}$$

$$48. (a) \quad A = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$$

eigenvalues

$$\lambda_1 = 5$$

$$\lambda_2 = -1$$

eigenvectors

$$\vec{k}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\vec{k}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \vec{x}(t) = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-t}$$

$$(b) \quad A = \begin{pmatrix} -1 & -4 \\ 1 & -1 \end{pmatrix}$$

eigenvalues

$$\lambda_1 = -1 + 2i$$

$$\lambda_2 = -1 - 2i$$

eigenvectors

$$\vec{k}_1 = \begin{bmatrix} 2i \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + i \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\vec{k}_2 = \begin{bmatrix} -2i \\ 1 \end{bmatrix}$$

$$\vec{x}_1 = e^{-t} \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \cos(2t) - \begin{bmatrix} 2 \\ 0 \end{bmatrix} \sin(2t) \right) = \begin{bmatrix} -2 \sin(2t) \\ \cos(2t) \end{bmatrix} e^{-t}$$

$$\vec{x}_2 = e^{-t} \left(\begin{bmatrix} 2 \\ 0 \end{bmatrix} \cos(2t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \sin(2t) \right) = \begin{bmatrix} 2 \cos(2t) \\ \sin(2t) \end{bmatrix} e^{-t}$$

$$\vec{x}(t) = c_1 \vec{x}_1(t) + c_2 \vec{x}_2(t)$$

$$\boxed{\vec{x}(t) = c_1 \begin{bmatrix} -2 \sin(2t) \\ \cos(2t) \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 2 \cos(2t) \\ \sin(2t) \end{bmatrix} e^{-t}}$$

$$(c) \quad A = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$$

$\lambda = 1$ is the only eigenvalue.

$\vec{k} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is an eigenvector

Let's find $\vec{p}$ so that $(A - \lambda I)\vec{p} = \vec{k}$

$$(A - I)\vec{p} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix} \vec{p} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\vec{p} = \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} \rightarrow \begin{cases} 2p_1 - 4p_2 = 2 \\ p_1 - 2p_2 = 1 \end{cases}$$

$p_1 = 1$ and $p_2 = 0$ is a solution.

$$\vec{p} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\vec{x}(t) = c_1 \vec{k} e^{\lambda t} + c_2 \left[\vec{k}t + \vec{p} \right] e^{\lambda t}$$

$$\vec{x}(t) = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^t + c_2 \left[\begin{bmatrix} 2 \\ 1 \end{bmatrix} t + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right] e^t$$

$$\boxed{\vec{x}(t) = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 2t+1 \\ t \end{bmatrix} e^t}$$

49.

$$A = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix}$$

eigenvalues

$$\lambda_1 = 4$$

$$\lambda_2 = 2$$

eigenvectors

$$\vec{K}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\vec{K}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\vec{X}(t) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{2t}$$

$$\vec{X}(0) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{cases} c_1 + c_2 = 2 \\ c_1 + 3c_2 = 1 \end{cases}$$

Solve the system to get: $\begin{cases} c_1 = 5/2 \\ c_2 = -1/2 \end{cases}$

$$\Rightarrow \boxed{\vec{X}(t) = \frac{5}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{4t} + \left(-\frac{1}{2}\right) \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{2t}}$$

$$\text{or } \vec{X}(t) = \frac{1}{2} \begin{bmatrix} 5e^{4t} - e^{2t} \\ 5e^{4t} - 3e^{2t} \end{bmatrix}$$

$$50. \quad \begin{cases} \frac{dx}{dt} = 2y + e^t \\ \frac{dy}{dt} = 3y - x - e^t \end{cases}$$

Matrix form

$$\frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 2 \\ -1 & 3 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\vec{X}} + \underbrace{\begin{bmatrix} e^t \\ -e^t \end{bmatrix}}_{\vec{F}}$$

Step 1 Find $\vec{X}_c$

eigenvalues

$$\lambda_1 = 1$$

$$\lambda_2 = 2$$

eigenvectors

$$\vec{K}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\vec{K}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\vec{X}_c = c_1 \overbrace{\begin{bmatrix} 2 \\ 1 \end{bmatrix}}^{\vec{x}_1} e^t + c_2 \overbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}^{\vec{x}_2} e^{2t}$$

Step 2 Find $\vec{X}_p$

$$\Phi = \left[\vec{x}_1 \mid \vec{x}_2 \right] = \begin{bmatrix} 2e^t & e^{2t} \\ e^t & e^{2t} \end{bmatrix}$$

$$\vec{x}_p = \Phi \int \Phi^{-1} \vec{F} dt$$

$$\det(\Phi) = e^{3t} \rightarrow \Phi^{-1} = \frac{1}{e^{3t}} \begin{bmatrix} e^{2t} & -e^{2t} \\ -e^t & 2e^t \end{bmatrix}$$

$$\Phi^{-1} = \begin{bmatrix} e^{-t} & -e^{-t} \\ -e^{-2t} & 2e^{-2t} \end{bmatrix}$$

$$\Phi^{-1} \vec{F} = \begin{bmatrix} e^{-t} & -e^{-t} \\ -e^{-2t} & 2e^{-2t} \end{bmatrix} \begin{bmatrix} e^t \\ -e^t \end{bmatrix} = \begin{bmatrix} 2 \\ -3e^{-t} \end{bmatrix}$$

$$\int \Phi^{-1} \vec{F} dt = \begin{bmatrix} 2t \\ 3e^{-t} \end{bmatrix}$$

$$\vec{x}_p = \begin{bmatrix} 2e^t & e^{2t} \\ e^t & e^{2t} \end{bmatrix} \begin{bmatrix} 2t \\ 3e^{-t} \end{bmatrix} = \begin{bmatrix} 4te^t + 3e^t \\ 2te^t + 3e^t \end{bmatrix}$$

$$\vec{x} = \vec{x}_c + \vec{x}_p$$

$$\boxed{\vec{x} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t} + \begin{bmatrix} 4t+3 \\ 2t+3 \end{bmatrix} e^t}$$

$$\Rightarrow \begin{cases} x = 2c_1 e^t + c_2 e^{2t} + (4t+3)e^t \\ y = c_1 e^t + c_2 e^{2t} + (2t+3)e^t \end{cases}$$

$$51. \quad \vec{x}'(t) = \underbrace{\begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}}_A \vec{x}(t) + \underbrace{\begin{bmatrix} 2 \\ t \end{bmatrix}}_{\vec{F}}$$

$$\vec{x}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Step 1 Find $\vec{x}_c$

eigenvalues

$$\lambda_1 = 1$$

$$\lambda_2 = -1$$

eigenvectors

$$\vec{k}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\vec{k}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\vec{x}_c = c_1 \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\vec{x}_1} e^t + c_2 \underbrace{\begin{bmatrix} 1 \\ 3 \end{bmatrix}}_{\vec{x}_2} e^{-t}$$

Step 2 Find $\vec{x}_p$

$$\vec{x}_p = \Phi \int \Phi^{-1} \vec{F} dt$$

$$\Phi = \begin{bmatrix} e^t & e^{-t} \\ e^t & 3e^{-t} \end{bmatrix}$$

$$\det(\Phi) = 2 \rightarrow \Phi^{-1} = \frac{1}{2} \begin{bmatrix} 3e^{-t} & -e^{-t} \\ -e^t & e^t \end{bmatrix}$$

$$\begin{aligned} \Phi^{-1} \vec{F} &= \frac{1}{2} \begin{bmatrix} 3e^{-t} & -e^{-t} \\ -e^t & e^t \end{bmatrix} \begin{bmatrix} 2 \\ t \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 6e^{-t} - te^{-t} \\ -2e^t + te^t \end{bmatrix} \end{aligned}$$

$$\int te^{-t} dt = -te^{-t} - e^{-t}$$

$$\int te^t dt = te^t - e^t$$

$$\begin{aligned} \Rightarrow \int \Phi^{-1} \vec{F} dt &= \frac{1}{2} \begin{bmatrix} -6e^{-t} + te^{-t} + e^{-t} \\ -2e^t + te^t - e^t \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} -5e^{-t} + te^{-t} \\ -3e^t + te^t \end{bmatrix} \end{aligned}$$

$$\vec{x}_p = \begin{bmatrix} e^t & e^{-t} \\ e^t & 3e^{-t} \end{bmatrix} \cdot \frac{1}{2} \begin{bmatrix} -5e^{-t} + te^{-t} \\ -3e^t + te^t \end{bmatrix}$$

$$\vec{x}_p = \frac{1}{2} \begin{bmatrix} -5 + t & -3 + t \\ -5 + t & -9 + 3t \end{bmatrix} = \begin{bmatrix} t - 4 \\ 2t - 7 \end{bmatrix}$$

Step 3 Find c_1, c_2

$$\vec{x} = \vec{x}_c + \vec{x}_p$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t} + \begin{bmatrix} t - 4 \\ 2t - 7 \end{bmatrix}$$

$$\text{since : } \vec{x}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{then : } c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + \begin{bmatrix} -4 \\ -7 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{cases} c_1 + c_2 = 4 \\ c_1 + 3c_2 = 8 \end{cases} \xrightarrow{\text{solve}} \begin{cases} c_1 = 2 \\ c_2 = 2 \end{cases}$$

$$\therefore \vec{x} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + 2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{-t} + \begin{bmatrix} t - 4 \\ 2t - 7 \end{bmatrix}$$

or

$$\vec{x} = \begin{bmatrix} 2e^t + 2e^{-t} + t - 4 \\ 2e^t + 6e^{-t} + 2t - 7 \end{bmatrix}$$
